Ziqing Guo

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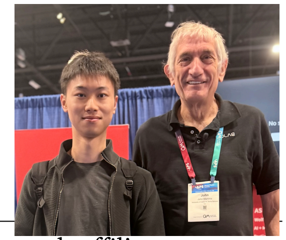

Profile

Hi:) I am a PhD candidate at Texas Tech University and a Quantum Error Correction research affiliate at Lawrence Berkeley National Lab (NERSC), specializing in GPU-accelerated quantum circuit simulation and quantum-HPC workflows. I design, build, and scale high-performance simulation libraries in **C++** and **Python**, using **CUDA-Q** and NVIDIA **cuQuantum** (**cuTensorNet**, **cuStateVec**) with multi-node **MPI** on the 256-node Perlmutter supercomputer, and I develop **tensor-network (MPS)** methods for large-scale circuit decomposition and cutting. My research spans arithmetic quantum algorithms and block encoding, quantum error correction (surface codes & QLDPC, Stim stabilizers), and agentic LLM systems for quantum circuit synthesis. My strengths lie in hardware-aware performance optimization, experimental and mathematical rigor, coding, and cross-disciplinary collaboration. My mentors are Jan Balewski (CA), Alex Khan (MD, DC), Steven Rayan (Usask), and Ziwen Pan (TX, AZ).

Keywords: GPU-accelerated quantum simulation (**CUDA-Q**, NVIDIA **cuQuantum**: cuTensorNet & cuStateVec), tensor networks (MPS circuit cutting), distributed HPC (multi-node MPI, Perlmutter), quantum error correction (surface code & QLDPC), quantum information (encoding), computing (QAOA).

Education

Texas Tech University, **PhD**, **High Performance Computing Center Fellow** — July 2026 (expected)

Newcastle University, **MSc**, **Advanced Computer Science, Merit** — Aug 2023

University of Tennessee, **Chengdu University of Information Technology**, **BE**, **Distinguished Graduate** — Jul 2021

Peer-reviewed Publications

- Ziqing Guo, Ziwen Pan. (2026). Stop quantum machine learning; do AI for quantum instead. Submitted to position paper of Conference on Neural Information Processing Systems (NeurIPS).
- Ziqing Guo, Ziwen Pan. (2026). RubriQ: Rubric-Guided Group Relative Policy Optimization for Constraint-Aware Quantum Circuit Synthesis. Submitted to International Conference for High Performance Computing, Networking, Storage, and Analysis (SC).
- Ziqing Guo, Jie Li, Yong Chen, Ziwen Pan. (2026). NNQA: Neural-Native Quantum Arithmetic for End-to-End Polynomial Synthesis. Submitted to International Conference on Machine Learning (ICML).
- Ziqing Guo, Ziwen Pan. (2026). Amortized Circuit Synthesis: Fast Parameter Inference for Quantum Polynomial Root Finding (APS March Meeting). Accepted.
- Ziqing Guo, Jan Balewski, Wenshuo Hu, Alex Khan, Ziwen Pan. (2025). Quantum Approximate Walk Algorithm (Nature npj unconventional computing Submitted, APS March Meeting Accepted).
- Ziqing Guo, Jan Balewski, Ziwen Pan. (2025). ShardQ: Circuit Cutting and 3D Tensor Recomposition for Quantum Simulation on Superconducting Qubits. Submitted to IEEE International Conference on Quantum Computing and Engineering (QCE).
- Ziqing Guo, Jan Balewski, Ziwen Pan. (2025). Vectorized similarity attention with learnable encoding for quantum transformer. Accepted to the Association for the advancement of Artificial Intelligence quantum symposium (AAAI). Oral presentation.
- Ziqing Guo, Alex Khan, Victor S. Sheng, Shabnam Jabeen, Ziwen Pan. (2025). Quantum parallel information exchange (QPIE) hybrid network with transfer learning. In IOP Quantum Science and Technology. HTML
- Ziqing Guo, Steven Rayan, Wenshuo Hu, Ziwen Pan. (2025). [QAOA paper] Direct entanglement ansatz learning (DEAL) with ZNE on error-prone superconducting qubits. Accepted in IEEE International Conference on Quantum Computing and Engineering (QCE) Public Talk. PDF Oral presentation.

- Ziqing Guo, Jan Balewski, Ziwen Pan. (2025). Q-GEAR: Improving quantum simulation framework. In 54th International Conference on Parallel Processing (ICPP). Oral presentation. Accepted. Best Poster Award PDF

Employment Experience

Research Applied Scientist, Leadership, Lightrider Inc — May 2026 – Aug 2026

Quantum Error Correction Research Affiliate, Lawrence Berkeley National Lab, NERSC — Jun 2024 – Present

Research Fellow, Texas Tech University — Sep 2023 – Present

Research Assistant, Newcastle University — Jun 2022 – Jun 2023

Cloud Engineering Intern, CISCO — Dec 2021 – Jun 2022

Grant & Awards

- **WCoE Grant Texas Tech University** $500 — Mar 2026
- **AAAI Quantum Symposia** Travel Grant $850 — Nov 2025
- **Texas Quantum Summit** Travel Grant $1k — Oct 2025
- **NSF** ICPP Travel Grant $1k — Sep 2025
- **Monterey Conference** Whiteacre College of Engineering Research Grant $700 — Sep 2025
- **IonQ Research Grant**, $350k — Jul 2025
- **IBM LBNL QCAN Award**, Contract, NERSC, DoE (No. DE-AC02-05CH11231) — Mar 2025
- **GenQ Quantum Hackathon**, $2.5k, Cat Qubit, First Prize — Oct 2024
- **Qiskit Quantum Summer School / Quantum Challenge**, Full Achievement — Jun 2024
- **AWS Braket Quantum Application Development**, Certificate — Mar 2024
- **AWS Braket Research Grant**, $2k, SV1, TN1 — Feb 2024
- **Pennylane Open Hackathon QHack / Code Camp**, Top Completionist — Jan 2024

Invited Talks

APS March Meeting, 2026
Special topic on Q-Gear generalization NVIDIA, 2025
International Conference on Parallel Processing, Quantum Computing, Sep 2025
International Conference on Quantum Computing, QCE, Aug 2025
Monterey Data Conference, Aug 2025
IBM Quantum / AI, TTU, Apr 2025
Improving quantum computation model, WCOE, Apr 2025
HackTX, University of Austin, Jan 2025 (Mentor)
Wave Technology, City of Calgary, Nov 2024
Platform Calgary, University of Saskatchewan, QAI Venture, Oct 2024
Berkeley National Lab, National Energy Research Computing, Quantum Group, Jul 2024
QuEra - NERSC quantum group neutral atom pattern formulation, Jun 2024
NVIDIA CUDA Quantum, QCAN, Jun 2024

Professional Services

npj Quantum Information
IOP Quantum Science and Technology
IEEE International Conference on Quantum Computing and Engineering
IEEE Transaction on Quantum Engineering
ACM Proceedings of the International Conference on Parallel Processing
Quantum and Beyond NEWSLETTER
Wolfram Research Student Ambassador
IEEE, ACM, APS, AAAI member

Projects

Q-GEAR: Hardware-Agnostic Quantum Circuit Simulation Framework github

- Developed a containerized simulation framework for large-scale quantum circuits on HPC systems like Perlmutter using Podman and SLURM.
- Integrated CUDA-accelerated kernels and PennyLane support to optimize high-performance quantum-classical workflows.

RubriQ: LLM-Guided Reinforcement Learning for Circuit Synthesis github

- Engineered a synthesis framework using Group Relative Policy Optimisation (GRPO) for automated, constraint-aware quantum circuit generation.
- Implemented rubric-guided rewards to optimize circuit depth and gate fidelity, aligning LLM outputs with physical hardware constraints.

MPS Based Quantum Circuit Decomposition and Cutting github

- Developed a circuit-cutting tool utilizing Matrix Product States (MPS) for efficient decomposition and recombination of large-scale circuits.
- Enabled the simulation of complex quantum systems across distributed environments by minimizing entanglement-induced communication overhead.

Quantum Parallel Information Exchange via Transfer Learning github

- Designed a hybrid communication framework leveraging mid-circuit measurements and transfer learning for optimized quantum data processing.
- Improved state-transfer efficiency in parallel quantum networks through adaptive circuit parameters and learned data encodings.

NNQA: Neural-Native Quantum Arithmetic Framework github

- Built a framework for translating neural network basis functions into native quantum arithmetic via polynomial transformation.
- Optimized high-dimensional data encoding layers to reduce circuit depth for AI to quantum applications.

Direct Parameter Augmentation for QAOA Optimization github

- Developed a parameter augmentation strategy for the Quantum Approximate Optimization Algorithm (QAOA) to solve complex QUBO problems.
- Implemented distributed solvers that enhance convergence rates for combinatorial optimization on near-term quantum devices.

Skills

Quantum (Proficiency): **Qiskit** (Heron-3 type superconducting), **CUDA-Q** (cuTensorNet, cuStatevector, NVIDIA-GPU), **IonQ** (Trapped Ion Aria-2, Forte), **PennyLane** (software stack), **Amazon Braket** (QaaS), **QCI-Dirac3** (Photonic QPU), **Fire Opal (Q-Ctrl)**, **TensorCircuit** (Tecent), **CirQ** (Azure, Microsoft)

Engineering: Python, NodeJs, NextJs, Mathematica, CUDA/MPI, Bash, Julia, Matlab, Cray HPC, Slurm, Container, DevOps, Scrapy/Data Mining, OpenRouter, Cursor, Automation, SQL, Cloud, Edge Computing, IaC, C++, Fortran, Rust, Go, Scipy, Yfinance, PyPortfolioOpt

LLM (Proficiency): **DeepSpeed** (ZeRO-1/2/3/Infinity, Offload strategies), **PyTorch** (FSDP, Distributed Data Parallel), **vLLM** (PagedAttention, High-throughput Serving), **LangGraph** (Stateful Multi-agent Orchestration), **Axolotl** (PEFT: LoRA, QLoRA Fine-tuning), **DSPy** (Declarative Prompt Programming), **LlamaIndex** (Agentic RAG, Context Management), **SGLang** (Structured Decoding), **TensorRT-LLM** (HPC Inference Optimization)

Interests: Guitar fingerpicker, Table tennis (competitive, shakehand grip), Calisthenics, Rollerblading, Skiing, Culinary.

Languages: English (proficient), Mandarin (native), Japanese (Elementary)